**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 078 · Lesson 078 · Binary Search: Bounds and Sorted Data

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Binary search precondition?

Q2. What does low represent?

Q3. What is mid for bounds low0 high6?

Q4. Binary search target15 in [1,3,5,7,9,11,13,15]: give midpoint indexes.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

low — Lowest candidate index.

high — Highest candidate index.

midpoint — Middle candidate.

halving — Repeatedly reducing interval by about half.

## New learning

Understand — Use binary search only when elements are sorted; keep inclusive low and high candidate bounds.

Trace — At every mismatch discard the midpoint and half the candidates with low=mid+1 or high=mid−1.

Explain — Each unsuccessful step shrinks the search interval. Worst time O(log n), iterative extra memory O(1).

## Worked example

Find14 in [2,5,8,11,14,17,20].

1. Check the working — mid3=11, mid5=17, mid4=14, returns index4.

# We Do · Practise with guidance

Find4 in [1,3,5,7,9] and explain absence.

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mid2=5, mid0=1, mid1=3, low2>high1: return−1.
```

Check the working

mid2=5, mid0=1, mid1=3, low2>high1: return−1.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Trace target11 in [1,3,5,7,9,11,13]. [6]

### Question 2

Write three binary-search comparison cases and correct bounds. [6]

### Question 3

Compare iterative binary and linear search with respect to time and precondition. [6]

# Exit Ticket · Answer before checking

After a too-small midpoint, setting low=mid always ensures progress.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Sorted data.

Q2. First possible candidate index.

Q3. Index3.

Q4. mid3=7, mid5=11, mid6=13, mid7=15; found index7.

### You Do pages · Independent questions

Q1. mid3=7 [2], low4 [1], mid5=11 [2], return5 [1].

Q2. equal return mid [2], smaller array mid→low=mid+1 [2], larger→high=mid−1 [2].

Q3. Linear O(n) [2] unsorted okay [1]; binary O(log n) [2] requires sorted data [1].

### Exit Ticket · Model answer

False. Use mid+1.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**